ActInf GuestStream 049.1 ~ "Clickbait, consciousness science, and respon

GuestStream | 59:06

Hello and welcome. It's July 26th, 2023. We're here in Active Inference Livestream 49.1 with Megan Peters and Nora Bradford. So thank you both for joining. We're looking forward to a presentation followed by a discussion on clickbait, consciousness science, and responsible journalism. So Megan, to you. Thank you. And thanks so much, Daniel, for having us. We're really happy to be here and to chat about this. So I'll talk for a little over 10 minutes and then we'll hopefully have a nice discussion here. So yeah, we're really happy to be here because I think that this is a really topical, really, yeah, very topical thing for us to be talking about right now. It's very in the media. It's really in response to the way the media has been covering consciousness science just in the past few weeks. So here we go. I'll have a few things to share with you. Okay, so since we're talking about consciousness science, let's start with a provocative question. So you are conscious. How is this possible? How is it possible that your brain creates the conscious experience of you? And so this is a captivating and it's a stimulating, kind of crazy question. It's a really huge question. It borders on the religious and spiritual in many ways. It borders on the cutting edge of physics in many ways. And so the study of this question and the field of consciousness science more generally, it's really straddling the border between the natural sciences and philosophy and spirituality in a way that a lot of other domains maybe don't need to do. And so as a result, this field of consciousness science, it really captures the hearts and minds and sometimes pretty wild imaginations of all sorts of folks, as I said, from psychologists and biologists and neuroscientists all the way to computer scientists and engineers, to religious and spiritual leaders, basically anybody who wonders about our origins or place in the universe, why that we know that we are here, why there is something that it is like to be us. They are the people who are interested in this question. And that's probably a lot of people who are listening and watching right now. So it's actually really quite easy, I think, to get swept up in the sensational nature of this question. And consciousness science is a really big deal. It touches on so many aspects of our everyday experience. And so this kind of leads us to another version of this question, which is, is this even really science? Is this something that can be even studied scientifically or does it transcend beyond scientific inquiry? Do consciousness scientists do something that can be characterized really as science? Or maybe they're studying something that's a little bit more on the border, like, I don't know, telepathy or panpsychism, like maybe rocks and trees are also conscious. Wouldn't that be crazy? Or maybe the vibrations of the universe mean that the universe is conscious. So it kind of quickly starts to feel a little fuzzy, a little less scientific. And that's what I want to talk about today is that perception and how that is kind of exacerbated by the way that this field is portrayed in the media and maybe what we can do about it. Okay. And so all this means that the consciousness science is really difficult, not just to do scientifically, and it is really difficult, I promise you. But it's also really difficult from a social perspective. So more than for some other

fields, those of us who do consciousness science and who study this scientifically, we have a really crucial and difficult responsibility, which is to accurately and precisely communicate the state of our field and its methods and its discoveries, and how it relates to other fields from physics to medicine and and beyond. And it's in fact, so difficult that our field has actually written a number of papers about just how difficult it is. So this is a fairly recent paper, TSP, including many others. And this is just from a few years ago. And these papers, and this one in particular, really highlight why our field and in specific has kind of an uphill battle compared to a lot of other areas of study, especially when it comes to potential misrepresentation in the media. So I thought that maybe in order to drive this point home, I can actually give you some examples of how consciousness science has been misrepresented in the media, so that we can get a handle on this, we can see it concretely. And as I'm giving you some of these examples, I want you to try to imagine how it would sound if the same kinds of statements, the same kind of sweeping claims were made about discoveries in biology or chemistry, neuroscience, climate science, engineering, that kind of thing. All right, so let's do some examples. Here is the first one. Okay, and this is an example which is actually 13 years old at this point, as well as I think you'll see maybe it hasn't gotten so much better over the past 13 years. But in this example, the title, Sizing of Consciousness by Its Bits, we have Christophe Koch, who is an expert on consciousness science at Caltech being quoted as saying, I love his ideas. It's the only real promising fundamental theory of... And here, it's deciding to present just this one scientist's opinion as fact, without challenging it at all, without presenting any other opinions at all. I'll point out that this opinion was actually not representative of the consensus in the field at that time, it was this one scientist's point of view, and it's not actually representative of the consensus in the field now either. And so I want you to take this quote, but I want you to replace consciousness with curing cancer. So this opinion presented as fact is akin to writing that one scientist's theory about a cure for cancer is the only really promising fundamental theory for curing cancer. And without presenting any other possibilities that maybe more work needs to be done or more experiments need to be run, this is really hyperbolic. And it seems like maybe this kind of hyperbole would be hopefully at least a little less likely to happen in biomedicine or at least would receive more pushback if it were presented this way.
So here's another example, and this one's a bit more recent. So this one's from 2018.
Okay, so the electrical activity that moves across neurons is thought by some scientists to contribute to human consciousness. This is like a perfectly reasonable statement, right? Like totally fine, totally legit. But then the journalist goes on to say, if electrical activity is being disrupted by anesthetic and plants too, causing them to lose consciousness, does that mean in some way that they are conscious? Okay, so first of all, these statements are pretty vague. That's not really super surprising from a popular science piece. And it's not really the primary problem here saying that electricity that moves across neurons is thought to

from disruption of electrical activity to plants being conscious. And this is pretty sensational, kind of to the extreme. And it makes this fuzzy border between consciousness science and pseudoscience, which is this border that we are trying so hard to delay all the time. It makes it even fuzzier, gives us an even harder problem to fight against as consciousness.

And if it isn't just happening 13 years ago, five years ago, it's still

why we're talking about

just last month, there was this which was held in conjunction with the Association for the Scientific

Study of Consciousness Annual Meeting, which was in New York City.

And at this public event, Christophe Koch, who you can see here on the right, and Dave Chalmers, here's on the left,

they presented their 25 year old bet about whether the neural correlates of consciousness, the ones that go up and

down with consciousness goes up and down, would be found.

And during this public event, some new results were unveiled for the very first time from one of the ongoing

projects, which was designed to challenge two of the many theories about how consciousness arises in the brain.

And those two theories are called the global neuronal workspace theory and the integrated information

theory. And the hot off the press results that were shown in this event, they actually ended up challenging

both theories. They showed that the neural correlates of consciousness are still unknown, that neither theory

was really strongly supported by the data that were collected, and so on. But that's not the

story that's not exciting. And so that's not the story that the press told. And so here's what the media

actually said. Okay, so first, we're going to start with nature. Nature, the journalism side of this

flagship scientific journal, you'd hope that they would be pretty careful about their scientific journalism.

But instead, they write taglines like this, like Christophe wagered Dave 25 years ago, that researchers

would learn how the brain achieves consciousness by now. But the quest continues.

But this is actually not what the bet was about. The bet was about the neural correlates, not the causal

mechanisms, not the generative production of consciousness. And remember, as good scientists,

scientists are not the same thing. And there's always a relevant XKCD. And so I direct you to this one here.

But it gets worse. So here's what science had to say. Advocates of the losing idea are not conceding

yet. So again, neither idea lost, remember, or rather both of them lost. Neither one of them won.

Nobody lost. Nobody won. And so why would science say that one of them lost or is even losing? That seems a

little over the top. And so it kind of makes you wonder, are they like trying to drive clicks? Or what

what is the point here of sensationalizing quite to this degree? It seems like a bad idea. It seems

like oversimplification that actually changes the meaning of what was presented. So here's another

example. This is from the economist. This article largely reported on a different event that was actually

part of the ASSC like primary meeting, but it's still capitalized on the buzz that this public event, this

evening event created. And it opens with this really kind of terrible twist of the actual experimental design.

So the economist article says top of the bill was the announcement of the result of a so-called adversarial

collaboration between proponents of two hypotheses about the nature of consciousness. Again, not what the bet was

about. It wasn't about the nature of consciousness, but anyway. This involved running a series of experiments begun in 2020 to determine which if either of them is correct. But again, that's not what the collaboration was designed to do, right? In science, you can never prove anything. You can never show you're
correct. You can only show that you are not wrong or that you are wrong. That's it. That's what you get in science.
Okay. So moreover, all of these articles, the three that I've shown you and a number of others, they also contained essentially no discussion of whether these two theories, global neuronal workspace theory and integrated information theory, one of which was losing apparently, whether they were the only ones worth considering. So the articles didn't really discuss anything else, any other possible theories. In fact, the articles called these two theories leading by the media, but the truth is that they're leading only in number of publications, not in amount of scientific evidence supporting them. They just have a lot of publications out there. There's no discussion in these articles beyond a trivial passing mention that there are any other theories out there at all, and especially that they are not, whether they are viable or not, whether they're empirically supported. So here's some, this isn't all of the empirically supported theories of consciousness out there. There's a really nice article here by Anil Seth and Tim Bain that came out last year that discusses a bunch of these.
And there's a really nice article here by this group also saying they actually looked at the empirical support for different theories of consciousness. How many papers are out there that actually challenge versus support each theory? And so this landscape is the one that should be the one to evaluate all of the results that were presented at this event in. It means that even if one of the two theories that this one project was looking at, if one of them didn't make the cut, that doesn't mean the other one wins and is right. It just means that one of them may be lost and it didn't even lose, remember.
And in fact, so in addition to these really nice papers, there's also a whole series of these adversarial collaborations from the same funding agency as the one that was shown in this public event that are designed to actually pit these theories against each other in a pairwise fashion, in a like Thunderdome style for all you Mad Max fans out there. I love old school sci-fi.
And the idea here is that two theories enter and one theory leaves. One theory survives, but it doesn't win because there's another round coming. And so all of these nuances, they were just unfortunately missing from all the news coverage. And I understand that news likes to get people to read it and it likes to be sensational, but the missing aspects of these new, the missing nuance really changes the meaning of what was presented. And that's actually a really big problem here. And why is this such a problem?
You might ask, I'm getting myself all worked up here, right? So the answer to why I'm getting all worked up really comes back to what I presented in the beginning, which is that we consciousness scientists have this uphill battle. We constantly have to fight to legitimize our research program.
And importantly, this isn't just in the eyes of the public. So it's not just in the eyes of the people reading these articles who are not part of this scientific sphere. It's also in the eyes of the scientific community more generally, of our funding agencies, and maybe most importantly,

our close peers within adjacent fields.
So imagine for a second that you are on a hiring committee at a university and you read a candidate's description of their research program, and it is about consciousness science. Okay. They have pages and pages and several publications, and they maybe have a grant that's about consciousness science. But you, as the member of the hiring committee, just read this, this New York Times article about how plants probably have consciousness. And so maybe your eyes are now rolling out of your head, because you've been primed to think this isn't real. This is sensational. This isn't something that can be studied scientifically. You might as well be talking about doing research on telepathy. Is this really a colleague that you want? So here's another example. What if you're on a grant review committee, or you're reviewing a paper, and you just read these articles in science and in nature that heavily imply that there are only two real theories of consciousness that have any empirical support.
But the paper or grant that you're reviewing is not targeting either one of these theories.
So now maybe you see the problem here.
These journalism articles, they're not just impuriating. They're really actively damaging the legitimacy of our field. They're cutting us off at the knees in the eyes of our peers in terms of the financial support we can compete for from granting agencies, in terms of where we can publish our science. And by the way, that science is state of the art and cutting edge. We have a responsibility, though, as people who are doing this state of the art consciousness science, to try to work with the media so that the reporting can be more honest. So we can continue working there as well. This is a two-way street in some ways. But we also hope that folks who are listening right now will recognize this pattern the next time you read an article about consciousness science, that maybe those sensationalized claims are not exactly a reflection of the state of art in the field. Because here's the truth.
This is real consciousness science, modern consciousness science. It's a collection of basically every single state-of-the-art discipline and method of scientific inquiry that you can think of.
So my version of consciousness science that I do in my lab is a conglomeration of cognitive sciences and computational neuroscience and a little bit of computer science, artificial intelligence, machine learning thrown in. We also have heavy influences from philosophy and psychology and engineering and a little bit of meta-science. And we have neuroimaging, we have statistics.
And this is really a representation of consciousness science, not just in my research group, but the consciousness science being done worldwide by thousands of people. Consciousness science is right at the center, right at the heart, the very bleeding edge of our collective scientific understanding.
So remember when I put this slide up before, and I said that the study of this question straddles the border between natural sciences and philosophy in a way that a lot of other domains maybe don't need to do. And as a result, it captures our imagination. It gets out there, it gets into your brain and your mind and sticks there in a way that some other fields might not.
But maybe you can see through what I've talked about today, that the challenges with communicating the crux of this question and in doing the science itself, it actually makes the people who are doing this pretty brave, especially the younger trainees. When you think about it, they're fighting this real

uphill battle. So personally, me, I'm relatively safe. I have job security. If people don't take me seriously when I say I'm a consciousness scientist, that's their loss. That's fine. But the younger people, the trainees in the field, those who are trying to get jobs, those who are trying to push the envelope of what we know about our brains and our minds and our awareness of ourselves in this world, they've got a lot of misperceptions to overcome. So if you are excited and you want to know more about the modern, exciting field of consciousness science, I suggest that you head to the ASSC website. You can have a look at the whole science program. They've actually posted videos of the keynotes from this year. So that's pretty cool. You can go and see some of those. You can also check out the plan for the future. And of course, if I've inspired you even a little bit to find out about consciousness science yourself, despite my kind of grant here, I say, welcome, please come join us. There's a lot of people doing some really cool stuff, really amazing mind blowing stuff working to figure this out. So please come join us. Absolutely. The more the merrier. And so I think I will stop there and just say, if you've enjoyed this talk or anything that I said resonated with you, I've got a real short little piece here. It's about 800 words. So not a long homework assignment. You can go and check that out. So I think I'll stop and pass the mic back to Daniel and also welcome Nora Bradford, who is a doctoral student here at UC Irvine, who works in consciousness science, but who also is an accomplished science journalist, who's written articles in Scientific American, Science News, Discover and many more. So thanks very much.

Awesome.

Nora, maybe to you for a first reflection about what you see there or how you came to be here?

Yeah, sure. Yeah, so I was at ASSC and went to the event that was associated with ASSC, but not run by ASSC, just to be clear. And it was really exciting to see consciousness science kind of being celebrated in the way that it was. I had some issues with the event, but overall, like I was very excited that it was getting this like public attention, right? And then afterwards, I was again, like excited to see that people were writing about my field because that hasn't happened. Journalism, the things I'm like to do, like immediately kind of disappointed by wrong. Yeah, and I guess I started to think about kind of why conscious science is maybe a harder journalist to cover. And I think there is just probably a lot more controversy and a lot more room for mistakes to be made than perhaps in fields where the research is more clear cut or more easily interpretable. So yeah, I think there's a lot as journalists and also as researchers from this. Yeah, I'll, can I just really quickly follow up and say, I think that Nora, you're spot on by saying that it's, or by presenting the idea that it's the interpretation that's part of the problem here. It's not just that people are maybe getting facts wrong when they're writing about this field. It's that the interpretations are really nuanced. They're really difficult to get right. And it's really easy for an oversimplification to end up fundamentally changing the meaning in this space in a way that maybe doesn't occur quite as often in some other fields.

Yeah, that's really interesting to kind of disentangle the features that make this a unique, although also exemplary case of communicating challenging ideas. I mean, the first that presents itself to me is just how much is on the table. I mean, whose pain matters, whose experiences or what's

experiences even matter? What's the difference between a who and a what? This is the real question.
If something doesn't have an experience, then there's no feeling. But if something does have an experience, then it's a feeling and it's a morally culpable or something that we're responsible for treating in an ethical way. So it's like, whether or not it's always brought up in those articles, the stakes are so high. Whereas whether a given molecular compound does this molecular interaction is closer to the objective, whereas this is the ultimate subjective question seemingly.
Dr. Sure. Yeah, I agree with you definitely that the ethical stakes, the moral implications of this field are just staggering. And certainly, it's not unique in that way that the ethical and moral implications of sociological research or of climate science or of many other domains also have these staggering implications. I think one of the biggest challenges that we have is not just in the interpretation of objective data, but exactly as you've just said, that this is the ultimate subjectivity. That the thing that we are studying by its definition is not objectively observable.
And that is absolutely something that is, at least you can design measurement tools in a way that have different philosophical foundations in basically every other field. But by its very nature, by the definition of subjective experience, we cannot design a measurement tool that actually says, what is it like to be you?
That's kind of the whole problem. But your points also about the ethical implications in medicine, the difference between a who and a what, I think that that's also really important to bring up.
That here we are trying to, I presented this as like, we want to understand consciousness. And I never really defined consciousness. I just said like, here's consciousness and like, here's an umbrella term.
And you take that kind of how you want. But it's true that consciousness science spans all the way from, you know, wakefulness versus coma, or locked in syndrome or traumatic brain injury, anesthesia, all the way through to, is a baby conscious, is a fetus conscious, is an AI conscious, is a cockroach, all right, or your dog. Or, you know, 200 years ago, people might argue that children were less conscious than adults. And so it was okay to hit your children, because they wouldn't feel it as much or something.
And so there's, there's a lot of sociological and ethical and moral implications in everything that we're doing, which is why one of the reasons I've chosen in this field, too, just that it really scratches a lot of intellectual itch, but it also feels really important, especially right now, with the rise of machine intelligence, and with ongoing debates about animal, or fetal consciousness that really bear extreme impact on legislation, on, you know, moral decisions that we make as a people. So yeah, it's, it's a big thing that we're doing. I'm a little biased, but it is a big thing.
And I would add that even if researchers are sometimes good at explaining what they mean by consciousness in their research, like to a journalist, say, the journalists are not always necessarily great at communicating that to the public for their writing. And I get why that would be hard. It's kind of, it can be clunky, it can be like, awkward and unappealing to explain. But that is crucial, because people, readers can take it however they want, based on how they're interpreting the word consciousness. So I think that's super important to address if you're

writing about consciousness.

Yeah, it's really a field that's kind of a, a leading indicator, because, especially if approached in a scientific way, in our current setting, a scientific understanding of X is what shapes evidence-driven policy about X. So I guess this leads to what, what is or, or would or could be a science-driven approach. Like, how do we know it when we see it, when someone is approaching this question from a scientific perspective? What would, what would each of you say to this?

Well, I'm going to give kind of the stock answer, which is that you want to have objective measures that are valid and reliable. But the big one is falsifiability. And that's, you know, any student of science going back to grade school is going to learn a little bit about that, which is that if you've got a theory or you've got a hypothesis about how, let's say the brain generates consciousness, or even going back to just correlates, just like which bits of the brain are, are correlated in their activity with whether you're more conscious or less conscious. So let's keep it really like very basic.

You need to have it be falsifiable. You need to have to say, it's this area of the brain that will correlate and not that area of the brain or that, and the recognition that an absence of evidence is not evidence of absence is also really crucial to this idea, which is that just because I failed to measure a correspondence between the activity and the bit of the brain that I hypothesized and, you know, some measure of awareness that I get from self report doesn't mean that that area of the brain isn't doing the work. It just means that our machines are terrible, or that our measure sucks.

And honestly, despite the state of the art measurements that we do have, we have, you know, millimeter level precision in humans, we have cellular level precision in non human animals, we can record a lot of stuff.

And we have a lot of really fancy machine learning algorithms to help us extract what that stuff means, to go from super high dimensional noisy data down to something that we can maybe make the argument is interpretable.

Interpretable, yeah. But just because you didn't find what you were looking for doesn't mean that it's not there. It just means that you were maybe looking in the wrong place or that you were looking using the wrong tool.

Interpretable, and so these these kind of go hand in hand, because on the one hand, you want your hypothesis to be falsifiable.

So you aren't allowed to just say, oh, well, I just wasn't looking for it the right way. So I'm just going to revise how I'm looking and try again, and therefore I'm still right.

You need to design your experiments and your theory such that if you find a particular result, a positive result, then it actually disproves your theory as opposed to just failing to provide support for your theory.

So I think that those those those those and other kind of standard criteria that you might find in any STEM classroom that demonstrate the scientific method.

And those are probably the definitions that I would fall back on.

And I think that consciousness and science, broadly speaking, the stuff that is presented, the ASSC, it does all of that.

It absolutely does. It's the same caliber of computational neuroscience and neuroimaging and psychological psychophysics measurement and logical philosophy approaches that you would find at any other flagship conference in the in the world.

Yeah. And I would say if you're reading an article rather than like a journal paper, like if you're reading something in the in the press, the way to know if it's trustworthy or not.

Or one way is if it's making sensational claims like claiming to have found the root of consciousness or some really large claim.

It's probably not true. I think we would that wouldn't happen in one study, I'm assuming.

So just read with a skeptical eye. And if you feel comfortable, try to read the primary literature, like get inspired by the press.

But if you feel like there's something off, just go to the primary literature and see what's up and and see if you can interpret it for yourself.

It's understandable if you can. But yeah, that's what I would say.

That's a good rule of thumb, Nora, that like one study is never going to be the answer in any way.

Yeah, not just ours. But it's it's easy to get swept away in the sensational claims and to be like, ooh, they found like the neural cause of consciousness.

Like probably not. Yeah. Yeah. And I would say like pre registration also.

I mean, Megan, I don't know like what your rule of thumb is for this, but those I mean, if people pre register their, you know, plan for the study and how they're going to analyze it ahead of time, that is usually a really good sign.

I mean, basically what you're just saying, but just like stamp of we promise we did this out of time kind of thing.

Yeah.

What's harking?

And hypothesizing after the fact, right?

Yeah.

Yeah.

Yeah, it was kind of cool to see meta science and also maybe kind of a bibliographic analyses looking at different trends in literature, but then also going beyond the peer reviewed literature to looking at the real trends that influence people who are outside of the profession.

And it's like, um, it's kind of, I don't know if canary in the coal mine is the right bird in the right niche here, but it's like, um, proposing unobservables is commonplace in research, like latent cause analysis and any kind of structural equation modeling.

And here is the, um, the tour de force of unobservables, maybe that which is defined as unobservables.

Definitely.

And so it's actually like a space to, to bring up all these pieces of epistemic hygiene about from pre registration to the pace and the tone of the scientific community and, um, questions of like multidisciplinarity in research.

And so it's, um, it's, it's kind of, um, it's a fun, and then there's a heart at the intersection of the Venn diagram where it kind of all comes together.

It's like, well, this synthesis cannot happen without all of these pieces.

If we're not having philosophers in conversation with the statisticians, then this synthesis can't happen.

Now, maybe the synthesis that you care about doesn't need that, but we care about this and this synthesis requires it.

So we need to develop the approaches and the communication around that.

And, um, so it's, uh, really a tip of the spear effort, even apart from the consequences of the research.

What a heartwarming sentiment.

Thank you for sharing that.

I, I think I love the phrase epistemic hygiene, because that is so crucial, I think to, to every field, but it's like particularly aggressively exposed in this field, which is exactly why I think that, um, my science has, has benefited so powerfully from close collaboration with philosophers and with statisticians.

I mean, this is kind of the space that I live in here in the cognitive sciences department is where we're halfway between AI and philosophy and psychology and neuroscience and engineering.

And it's like, it's kind of like where all these things come together, but then this meta science thing, I'm happy that you picked up on that.

Um, cause I think that it's also really the, the bibliometric analyses are really critical and we're at this crazy spot and have been.

For a long time now, but I think it's really, really coming to a head where the vast majority of our scientific knowledge is not accessible to anyone researcher simply because there's so much of it.

Right.

But you can't read all of the literature.

You can't even read all the literature that's in your tiny little corner and consciousness science.

And so I think that's, like many other fields in this way, actually is one of those that brings together a lot of different fields.

So it makes it even harder for you to keep up with the things that are going to be really relevant.

I mean, you have to keep up with the cutting edge tools in neuroimaging data analysis, plus the epistemic hygiene that the philosophers are going to bring.

Plus like cutting edge behavioral paradigms and what they mean.

And like, you know, naturalistic data, uh, collection settings so that you're not just saying, Oh, I've discovered something fundamental about consciousness by showing you Gabor patches all day.

And I say this as someone who shows people Gabor patches all day, but it's, it's really like, it's a real challenge to bring together all these things.

And maybe you're right.

Maybe that consciousness sciences, I don't know about the canary in the cold mine, so to speak, but maybe you're right that it's one of the places where you can really see all these cogs working together to come at one major question, or maybe you don't quite see the need for that high.

And some other fields, yet it still is there.

Like latent cause analysis is not just something that we do in consciousness science or neuroscience.

We do it in economics.

We do it in sociology.

We do it like everywhere.

And so having a real handle on how to design stuff that actually ask the questions that you're interested in is something that I find that I have to continuously work hard to keep in my sights.

And I'm lucky that we've got a pile of philosophers that we get to hang out with who blow into your talk and say everything that you just said is wrong.

And here's why here's this logical hole in what you've done.

And they're right.

And it's great because it makes the science better.

And you don't typically get philosophers, you know, blowing into a talk and a lot of other fields, it doesn't happen.

And, and I think it should.

We all have that inroad to our experience.

So it is something that integrates across disciplines, kind of even cast the disciplinary organizational paradigm into at least critical light.

And, and, and then, um, goes even outside of academia because these are, these are what people search engine for and, uh, language model discuss.

And, and I think it is one of these, um, it's like an itch that some can't scratch with this inscrutable and ineffable nature that, that does verge into the mystical.

And then like, so what do we do with the huge mystery?

And so that, that, that makes it a, um, a, a, a training and exhibition fields.

Um, again, that's not just important, but like, is kind of the, the, um, the archetype of, well, how do you have a multi-decade conversation with potentially non-compatible?

Theorizing.

Whereas in many of the physical sciences, including physical reductionist perspectives on, on neuroscience, threads open and close a little bit quicker, a little bit easier, tend to not, um, delve into, uh, ancient history, tend to not need to invoke world knowledge traditions.

So that makes it fun.

It makes it fun and useful.

And if something's fun and useful, what, what, what secret third thing are we looking for?

Exactly.

Sounds like, uh, if you weren't already a consciousness scientist, maybe we've converted you a little bit.

Um, as, as only a bug doctor, I did, did, uh, one, one work on the, uh, ant colony and definitions of consciousness.

And we didn't come down on the issue of whether, like, the nest mates are conscious under what definition or the colony.

Um, but more pointed out, well, if you define it according to some neuro, mammalian neuroanatomy, then you kind of already know what the answer is going to be.

It's going to be, it's going to be no.

And if you define it according to a specific, um, primate centered symbolic communication approach, again, you know what that's going to be.

It's no.

If you define it based upon some generic informational property, then you already know what it's going to be.

It's going to be on that continuum.

And so we kind of tried to flip the approach by instead of asking like, which, um, feature would help us differentiate whether the ant colony is conscious or not.

We just said, let's just look at what the features that people are using and then ask whether colony is, but, but we don't have, we don't have the, um, one meter metal bar for consciousness.

So again, it's like, well, let's say seven out of 10 that we observed, we don't know how many other unproposed theories there are.

So seven out of 10 could be total, um, uh, sampling bias.

Um, even if it was 10 out of 10, then what would that type of consensus really mean?

And, um, that's where our road took us.

And I, I saw that conclusion as like a chance to broaden the discussion.

But discussions are not only about broadening because there's real decisions that are being made.

So it can't only be yes.

And, and I think that when you connect it back to falsification and all of these other properties, that's where we get the, the winnowing.

But it's those two complimentary conversations like the broader, um, inclusion of perspectives that expands the space that we can comprehend.

And then the diligence that, that also shrinks it back down and, and somehow will zigzag to better and better understandings.

So, um, your, uh, your point about, we don't have a yardstick.

We don't have a metal rod, right.

That says, you know, how conscious, I mean, like, according to, to some folks, we, we have that metal rod and we've surpassed that with, for example, large language models.

Uh, there, uh, and we're not just talking, you know, particular Google engineers who decide to get really excited about how their pet language model gets, is conscious.

We're talking any, any kind of reasonable interpretation of the old school Turing test would be passed by existing language models right now.

In fact, arguably it might have been passed long ago with Eliza, uh, in like the seventies.

Um, and, and if it wasn't passed with Eliza, then certainly it has been passed now.

But I think that there is a growing understanding that information processing capacity and intelligence are not the same thing as internal phenomenal experience.

And that the tests that we have to the extent that we have any tests at all are woefully inadequate.

And as you said, are locked to particular implementation.

So you can't apply the mammalian test for consciousness.

Like, Oh, I'm looking for this particular neural signature, which tells me whether the monkey is anesthetized or not.

And I can't apply that to ants and expect it to work.

It's not going to tell me anything useful.

There can't necessarily apply it to octopuses either for the record.

So we're not even going insects.

We're going other arguably incredibly intelligent beings that can open jars and be sneaky and like do all sorts of cool stuff.

And, uh, and that have complex social interactions and are basically aliens.

This is my, my, you know, professional scientific opinion here, but they basically are right.

Like they're so fundamentally physically different from us.

And so any consciousness or meter that we develop for us, or that applies in monkeys or even in dogs or cats is not really going to apply there.

Um, and so differentiating tests for intelligence from tests, from like inner experience, it's a really difficult problem.

And then there's the third problem, which is a test for whether we attribute consciousness to an agent, which if you saw the movie, Ex Machina is the Garland test.

Um, so that's like, you know, do nevermind whether it actually has consciousness.

Do I attribute consciousness to it?

Because that is going to change how we interact with it and the rights that we ascribe to it and so on.

And so I think we're well past the point in which we need to get a better handle on this.

Some of the stuff that I'm working on right now is moving in this direction, not just me and a huge group of lots of other people.

Um, and I think Yashua Bengio gave testimony to Congress like today, uh, about the need for, uh, increased regulatory oversight of, uh, who has access to so-called artificial intelligence models so that we can better understand.

How their implementation and their interaction with humans might exacerbate biases or cause all sorts of other problems.

Um, so you can actually go and read his, his testimony that he gave.

Uh, I don't remember where the link is.

I'm going to find it.

Um, but he had a summary and then he links to the testimony.

And I think that having folks like Yashua and other giants in the field, speaking out about the need for actually tackling this and not, this isn't just a philosopher's playground to talk about consciousness.

This is, this is, as you said, it's a testing bed for where the rest of science is going.

And it's also like very topical right now.

So what is that healthy conversation?

What does that, um, conversation look like in the case of the synthetic intelligences and language models and all of this?

Um, what, what does that, um, continuum look like between the actual measures and proxies that we use in research?

And, uh, I mean, what memes or themes or words that I guess by definition aren't popular today?

Like big data wasn't popular until it was.

So what, something like what has to come on the scene and come into the lab, do you think, to support the kind of, um, like work and knowledge that, that either of you would want to see?

This isn't an answer to your question, but it just reminded me of one thing that we need less of.

People writing about these interactions with AI that are like, my AI told me that it wanted to get married or like my AI, whatever.

Like that's not, it's not useful.

It's not news.

Stop reporting about it.

That's all I have to say.

Okay.

Go ahead, Megan.

No, I, I wanted you to, to take this too, because I think that this question is about communication in a way that is, you're, you're much more embedded in the science communication space than I am.

And so for, I think one possible way to go about this, that maybe you can speak to would be what kinds of, you know, you're, Daniel, you're talking about what do we need to bring into the lab?

And maybe communication tools is one of those or ways of communicating this science ways of, of, and to different audiences.

Um, and so maybe Nora, you've got some ideas given you've done so much in this space.

I don't know.

Yeah.

I mean, I guess I, what we've been talking about with people who study human consciousness, um, people who study consciousness and other entities as well.

Like just being clear about what you mean by consciousness.

Like, do you mean a subjective experience?

Do you mean, um, you know, basically as a light on or off, like, uh, do you mean awake versus asleep?

Um, just being really clear about what level you're talking at, I think is really helpful.

Again, when you're just like presenting your research, whether it's on Twitter or you're talking to a journalist, um, really being clear cut about that so that people can't make kind of extrapolations from your work that your work doesn't cover.

Um, I don't know what else.

I mean, yeah, I guess just being willing to also talk about your work.

I feel like a lot of data science feels pretty inaccessible to the general public.

Um, so finding ways to communicate what you do, um, and how you're conducting your work is helpful and being open about it.

Yeah.

There's a lot of jargon in our field too.

In any field there's jargon.

Um, but the, the oversimplification, this is going back to what we talked about at the beginning, the oversimplification of jargon in terms in our field seems particularly susceptible to, uh, creating a

situation where the facts haven't changed, but the interpretation has.

Uh, and so I think it's almost like sensitivity training for anybody who's in this space, like just kind of a, you know, a little, a tutorial that you can do.

That's like a two hour, you know, click through things.

It's like, here are all the ways in which communicating this type of science due to its nature is more challenging than you might anticipate.

And so my, my own involvement in, um, in a program run by the Canadian Institute for advanced research has given me a little bit of training and speaking with the media.

And I would never, never say that I am an expert in this at all.

Uh, I will only say that, uh, it was eyeopening and it was very useful in the clarity of language that you are used to using the precision of language that you're used to using.

Uh, close scientific sphere versus when you're speaking to a more general audience.

And when you're speaking to a journalist or with several journalists who are then going to be writing for a more general audience, it's a, it's a different mindset.

It's a different way of speaking.

And even just being aware of that, I think is really, really useful.

It's not that one is better than the other.

It's just that they are very different.

And then when we are training to be scientists, when we're doing a PhD in a field related to this, speaking to the general public is not a skill that gets pressed on.

It's not a skill that gets, it's part of like your professional development training.

Maybe you do like one practice talk somewhere, but it's not typically part of a general training protocol.

And so speaking to non-specialists, speaking to the media, speaking to high school audiences, speaking to your mom at the dinner table, like learning how to do this effectively can actually be really valuable and really rewarding and really powerful.

And again, I would never say that I am an expert at this, at this point.

I just like to think that I'm a little better at it than I otherwise would be.

Had I not had it drilled into me, how careful I need to be.

I would also say, don't be scared to like email journalists if you see that something's inaccurate or, you know, speak up like Megan has done about how things are portrayed inaccurately.

Because, yeah, as a journalist, like no one's ever done that to me, but it would, if something was inaccurate, I would want to know and I would want to fix it like ASAP.

So don't feel bad about emailing journalists. They want their pieces to be accurate as well. So that's what I would add.

That can add to the conversation.

Yeah.

It does make me feel a little bad about like, you know, calling out the pieces that I did in the presentation and the piece of writing that I did too.

But at the same time, like they put this in writing and this is wrong. And, and it's, I don't like making people upset and I don't like making people angry.

But it's out there in the world and it's not a million eyeballs and that's a problem. And so we're hoping it's your, your viewership is about a million, right, Daniel? So give or take.

And so we're.

Low billions.

Low billions.

Low ail

but it's like but you miss the point of communication which it's not about what you would think about it

it's about who the recipient would be so it's kind of like the theory and the practice of of um

situational awareness about different minds all of which are different than our own

some are potentially radically different than our own others might be more similar um so it's

there's really um maybe even a quintessential relationship between consciousness studies

and communication studies which provides like a really unique two-way street that that um the world

is swerving into that two-lane highway with these questions that are grabbing people's salience

and what what are your sort of um like closing thoughts or directions or what are some exciting

areas that that you'd like to see unfolding in the coming months and years

well i'll go first um yeah first of all i i think you're spot on by by saying uh

uh communicating consciousness science being about knowing about other minds i mean it's like

effective communication is theory of mind and theory of mind may have a lot to do with modeling other

minds and modeling your own mind and yes absolutely it's all it's all wrapped up so i think for me um

getting this right going forward or continuing to to work towards getting it right is a priority for me

uh and one of the the pieces scientifically that i'm working on that i hope to then communicate

effectively when we get it right are several of the pieces is i have a couple um of these really

wonderful large collaborative projects where we're working on uh everything from figuring out how you

would actually measure consciousness effectively or what it might even take to measure consciousness

effectively in other entities that are not necessarily like awake behaving healthy adult human beings

all the way to let's actually design some experiments to test several different theories

of consciousness let's design a new uh some of what norah's working on actually is designing new

measurement approaches to say well we need to actually like have a quantitative handle on the

things that we're talking about so this becomes objective it becomes falsifiable it becomes like

model we can do model arbitration so all these things i'm really excited about but overarching all of

those is this idea of going it's all connected right you you've talked about how conscious of science might be

this this leading edge or or wave front of how uh interdisciplinary science and meta science needs to be done

going forward as uh as we become even more theory driven in some of our uh scientific lines of inquiry than

we have been in the past and more interested in latent variables and causal discovery and so on and so forth

and as the science teams get bigger and more complicated and the jobs within those science teams become more

specialized and we have more and more need for interdisciplinary collaboration we're gonna have to change how we do team science and how we do credit assignment within team science in a way that i think might end up being kind of incompatible with the nice siloed cute departmental structure that we have in our universities and so a focus on not just tolerating but celebrating interdisciplinarity celebrating team science doing credit assignment correctly and you know saying first author isn't the only thing that matters that like it really matters if you wrote this gorgeous software package that facilitated a whole bunch of stuff like that's also a meaningful and powerful contribution even if you didn't get first author on that paper you still deserve a lot of credit so i think that there's a lot of shaking up that we need to do about how we do science going forward and i really like the idea of using consciousness science as a case study i think that's beautiful and thank you very much for making that lovely connection because it really locks together a lot of the things that i've been thinking about recently as well so there's my my parting comments

yeah i was also going to bring up the um these huge projects that you're part of and that other people are part of it's really exciting to see so many people working together in these huge teams i feel like it just makes science so much more efficient and more careful so that's very cool another thing i'm hoping is that one that journalists don't get scared of covering consciousness science from this i think like we'll learn from our mistakes and we'll move forward i i don't think we'll get scared but i hope that there's more coverage of this field um and i also hope that i mean something that i've noticed about science journalism is they really like to just have clear-cut findings because that's the simplest thing to communicate to people but i think we can as a society like learn to be comfortable with the ambiguity a little bit more um especially when it comes to consciousness science like we might have found something but we're not sure and that's okay and that's science like it's okay if your headline isn't snappy and clear it's important for it to be clear but snappy and i guess have one takeaway point it's okay if it doesn't have a takeaway point um yeah i think that's what i'm excited about new era of science journalism well it sounds like bigger teams bigger community more theory more data more philosophy

more ambiguity more ambiguity it'll be a great time more philosophy more ambiguity love it yes

thank you megan and norah for joining always um welcome back so till next time yeah thank you so much this has been really a pleasure thanks for having us thanks thank you bye

thank you